

Task 1:

```
using System.Diagnostics;
```

```
string filePath1 = "C:\\Users\\surya.p\\source\\repos\\Day4\\Day4\\TextFile1.txt";
```

```
string filePath2 = "C:\\Users\\surya.p\\source\\repos\\Day4\\Day4\\TextFile2.txt";
```

```
void ReadFile1()
```

```
{
    Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] File 1 Started");
    var lines = File.ReadAllLines(filePath1);
    foreach( string line in lines )
    {
        Console.WriteLine(line);
    }

    Thread.Sleep(10_000);
    Console.WriteLine("File 1 Completed");
}
```

```
void ReadFile2()
```

```
{
    Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] File 2 Started");
    var lines = File.ReadAllLines(filePath2);
    //Console.WriteLine(lines);

    foreach (string line in lines)
    {
        Console.WriteLine(line);
    }
    Console.WriteLine("File 2 Completed");
}
Thread task1 = new Thread(ReadFile1);
Thread task2 = new Thread(ReadFile2);
```

```
Stopwatch sw = new Stopwatch();
```

```
sw.Start();
```

```
task1.Start();
```

```
task2.Start();
```

```
task1.Join();
```

```
task2.Join();
sw.Stop();
Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] - Time Taken :
{sw.Elapsed.TotalSeconds}");
```

```
async Task ReadFile1Async()
{
    Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] File 1 Started");
    var lines = File.ReadAllLines(filePath1);
    foreach (string line in lines)
    {
        Console.WriteLine(line);
    }

    await Task.Delay(10_000);
    Console.WriteLine("File 1 Completed");
}
```

```
async Task ReadFile2Async()
{
    Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] File 2 Started");
    var lines = File.ReadAllLines(filePath2);
    //Console.WriteLine(lines);

    foreach (string line in lines)
    {
        Console.WriteLine(line);
    }
    Console.WriteLine("File 2 Completed");
}

sw = new Stopwatch();
Task Atask1 = ReadFile1Async();
Task Atask2 = ReadFile2Async();
```

```
sw.Start();
Task.WaitAll(Atask1,Atask2);
```

```
Console.WriteLine($"[{Thread.CurrentThread.ManagedThreadId}] - Time Taken :
{sw.Elapsed.TotalSeconds}");
```

```
[13] File 2 Started
[12] File 1 Started
The Text File 1 content
Sample Content in File 1
The Text File 2 content
Sample Content in File 2

File 2 Completed
File 1 Completed
[1] - Time Taken : 138.6331706
[1] File 1 Started
The Text File 1 content
Sample Content in File 1
[1] File 2 Started
The Text File 2 content
Sample Content in File 2

File 2 Completed
File 1 Completed
[1] - Time Taken : 10.0255509
```

Task 2:

```
Test Status : Completed
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace Day4
```

```
{
    internal class Student
    {
        public delegate void pass_info(string msg);
        public pass_info info;
        public Student(pass_info info)
        {
            this.info = info;
        }

        public void write_test()
        {
            info("Completed");
        }
    }
}
```

```
    }  
  }  
}
```

```
using System;  
using System.Collections.Generic;  
using System.Globalization;  
using System.Linq;  
using System.Text;  
using System.Threading.Tasks;
```

```
namespace Day4  
{  
    internal class Teacher  
    {  
        public Student stu;  
  
        public Teacher()  
        {  
            stu = new Student(test_completed);  
        }  
  
        public void test_completed(string msg)  
        {  
            Console.WriteLine($" Test Status : {msg}");  
        }  
    }  
}
```

```
using Day4;
```

```
Teacher teacher = new Teacher();
```

```
teacher.stu.write_test();
```